WORKED SOLUTION

STEP I 1991, Question 8

Pure • Integration by Substitution

STEP I 1991, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q8

The key idea

The substitution $y = \pi - x$ maps $[0, \pi]$ to itself and leaves $\sin x$ unchanged, but it turns the awkward factor x into $\pi - y$. Adding the integral to its reflected copy makes the x disappear altogether. In part (ii) the integrand is built from a function and its derivative, so one well-chosen substitution does all the work.

Solution

Part (i)

Write $I = \int_0^\pi x\,\mathrm{f}(\sin x)\,\mathrm{d}x$. We want a substitution that swaps the ends of $[0, \pi]$ and keeps $\sin x$ the same, so take $k = \pi$: put $x = \pi - y$, so that $\mathrm{d}x = -\mathrm{d}y$, $\sin x = \sin(\pi - y) = \sin y$, and $x = 0, \pi$ correspond to $y = \pi, 0$. Then

$$\begin{aligned} I &= \int_\pi^0 (\pi - y)\,\mathrm{f}(\sin y)\,(-\mathrm{d}y) = \int_0^\pi (\pi - y)\,\mathrm{f}(\sin y)\,\mathrm{d}y \\ &= \pi \int_0^\pi \mathrm{f}(\sin y)\,\mathrm{d}y - I, \end{aligned}$$

since y is only a dummy variable, so the last integral is I again. Hence

$$I = \frac{\pi}{2} \int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x.$$

? **Ask yourself** How do we get from $\frac{\pi}{2}\int_0^\pi$ to $\pi \int_0^{\pi/2}$?

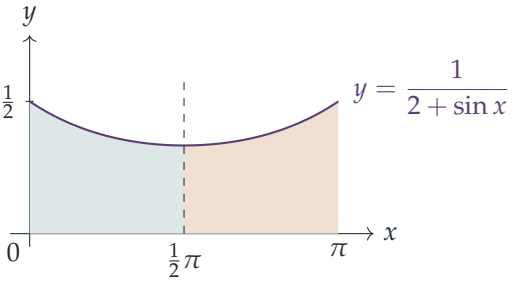
The graph of $\mathrm{f}(\sin x)$ on $[0, \pi]$ is symmetric about $x = \frac{1}{2}\pi$, because $\sin(\pi - x) = \sin x$. To make this precise, split the integral at $\frac{1}{2}\pi$ and use the same substitution $x = \pi - y$ on the second half:

$$\int_{\pi/2}^\pi \mathrm{f}(\sin x)\,\mathrm{d}x = \int_{\pi/2}^0 \mathrm{f}(\sin y)\,(-\mathrm{d}y) = \int_0^{\pi/2} \mathrm{f}(\sin y)\,\mathrm{d}y.$$

So $\int_0^{\pi} \mathrm{f}(\sin x)\,\mathrm{d}x = 2\int_0^{\pi/2} \mathrm{f}(\sin x)\,\mathrm{d}x$, and therefore

$$\int_0^{\pi} x\,\mathrm{f}(\sin x)\,\mathrm{d}x = \pi \int_0^{\frac{1}{2}\pi} \mathrm{f}(\sin x)\,\mathrm{d}x,$$

as required.



Now take $\mathrm{f}(u) = \dfrac{1}{2+u}$, which is continuous for $u \in [-1,1]$. Then

$$\int_0^{\pi} \frac{x}{2+\sin x}\,\mathrm{d}x = \pi \int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{2+\sin x}.$$

For the remaining integral use $t = \tan\frac{1}{2}x$, so that

$$\sin x = \frac{2t}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2},$$

and $x = 0, \frac{1}{2}\pi$ correspond to $t = 0, 1$. Then

$$\begin{aligned}
\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{2+\sin x} &= \int_0^1 \frac{1}{2+\frac{2t}{1+t^2}} \cdot \frac{2\,\mathrm{d}t}{1+t^2} = \int_0^1 \frac{2\,\mathrm{d}t}{2(1+t^2)+2t} = \int_0^1 \frac{\mathrm{d}t}{t^2+t+1} \\
&= \int_0^1 \frac{\mathrm{d}t}{\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}} = \frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2t+1}{\sqrt{3}}\right]_0^1 \\
&= \frac{2}{\sqrt{3}}\left(\tan^{-1}\sqrt{3} - \tan^{-1}\frac{1}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}}\left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \frac{\pi}{3\sqrt{3}}.
\end{aligned}$$

Hence

$$\int_0^{\pi} \frac{x}{2+\sin x}\,\mathrm{d}x = \frac{\pi^2}{3\sqrt{3}} = \frac{\sqrt{3}\,\pi^2}{9}.$$

Part (ii)

Ask yourself What is the derivative of $(\sin^{-1} t)^2$?

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\sin^{-1}t)^2 = \frac{2\sin^{-1}t}{\sqrt{1-t^2}},$$

which is (twice) exactly the factor multiplying the cosine. So substitute $u = (\sin^{-1}t)^2$, with $\mathrm{d}u = \dfrac{2\sin^{-1}t}{\sqrt{1-t^2}}\,\mathrm{d}t$. When $t = 0$, $u = 0$; when $t = 1$, $\sin^{-1}1 = \frac{1}{2}\pi$, so $u = \frac{1}{4}\pi^2$. Therefore

$$\int_0^1 \frac{(\sin^{-1}t)\cos[(\sin^{-1}t)^2]}{\sqrt{1-t^2}}\,\mathrm{d}t = \frac{1}{2}\int_0^{\pi^2/4}\cos u\,\mathrm{d}u = \frac{1}{2}\sin\frac{\pi^2}{4}.$$

The integrand blows up at $t = 1$, so strictly this is an improper integral. The substitution handles it cleanly: in terms of $\theta = \sin^{-1}t$ the integral is $\int_0^{\pi/2}\theta\cos\theta^2\,\mathrm{d}\theta$, whose integrand is continuous, so the integral converges and the value above is correct. (Doing it in two steps, $\theta = \sin^{-1}t$ then $u = \theta^2$, is a perfectly good alternative route.)

The answer $\frac{1}{2}\sin\frac{1}{4}\pi^2$ is exact and doesn't simplify further: $\frac{1}{4}\pi^2$ is not a 'nice' angle, which is presumably why the examiners warned against reaching for a calculator.

Checks

- Simpson's rule gives $\int_0^{\pi} x/(2+\sin x)\,\mathrm{d}x \approx 1.899406$, and $\pi^2/(3\sqrt{3}) \approx 1.899406$.
- A quick sanity check: $\frac{1}{3} \le \frac{1}{2+\sin x} \le \frac{1}{2}$, so the integral lies between $\frac{1}{3}\cdot\frac{1}{2}\pi^2 \approx 1.64$ and $\frac{1}{2}\cdot\frac{1}{2}\pi^2 \approx 2.47$.
- The identity in (i) can be tested with $\mathrm{f}(u) = 1$: both sides equal $\frac{1}{2}\pi^2$.
- For (ii), $\frac{1}{2}\sin(\frac{1}{4}\pi^2) \approx 0.312133$, which matches a numerical evaluation of $\int_0^{\pi/2}\theta\cos\theta^2\,\mathrm{d}\theta$.

Common slips

- Stopping at $I = \frac{\pi}{2}\int_0^{\pi}\mathrm{f}(\sin x)\,\mathrm{d}x$ without justifying the halving of the range.
- Forgetting to change the limits when substituting $t = \tan\frac{1}{2}x$ (the upper limit becomes 1, not ∞).
- Losing the factor $\frac{1}{2}$ in part (ii): $\mathrm{d}u$ carries a 2.

Beyond STEP

The trick in part (i), exploiting a symmetry of the interval to evaluate $\int x\,g(x)\,\mathrm{d}x$ without ever finding an antiderivative, is a small example of a big theme: symmetry often tells you the value of an integral more cheaply than calculation. Part (ii) touches the theory of improper integrals, where a change of variables can turn an unbounded

integrand into a continuous one. Abbott's *Understanding Analysis* makes the change-of-variables theorem rigorous, and Siklos's *Advanced Problems in Mathematics* has many more integrals of this type.

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